Methodology:

Part A — Data Preparation

- Loaded and cleaned both datasets.
- Converted timestamps to datetime format.
- Merged trader data with sentiment data by date.
- Created key metrics:
    - Daily PnL per trader
    - Win rate
    - Average trade size
    - Number of trades per day
    - Long/Short ratio
    - PnL volatility (risk proxy)
- Created aggregated dataset grouped by sentiment.

Part B — Analysis

The following analyses were performed:

- Performance comparison (PnL, Win Rate) across sentiment regimes.
- Risk analysis using drawdown proxy and maximum daily loss.
- Behavioral analysis (trade frequency, trade size, long/short bias).
- Trader segmentation:
    - High vs Low Frequency traders
    - High vs Low Position Size traders
    - Consistent vs Inconsistent traders

Part C — Strategy Recommendations

Based on the analysis, sentiment-based trading strategies and risk management rules were proposed.

Insights:

1. **Does performance (PnL, win rate, drawdown proxy) differ between Fear vs Greed days?**

   Trader profitability is higher during Fear and Extreme Fear market conditions compared to Greed periods. While Extreme Greed shows occasional large profits, the median profitability is lower and more volatile. This suggests that traders perform more consistently during Fear periods, possibly due to more cautious and disciplined trading behavior.

Win rates are highest during Extreme Greed periods; however, this does not translate into the highest profitability. This indicates that traders may be taking frequent small-profit trades during Greed periods but are exposed to occasional large losses, reducing overall profitability.

Risk levels, measured using PnL volatility and maximum daily loss, vary across sentiment regimes. Although Fear periods show higher volatility, the largest single-day losses occur during Greed periods. This suggests that traders take excessive risk during Greed markets, leading to larger drawdowns despite lower overall profitability.

Thus, we can conclude that trader performance differs significantly between Fear and Greed market conditions. Fear and Extreme Fear periods are associated with higher median profitability and more consistent returns. In contrast, Greed and Extreme Greed periods show higher win rates but lower overall profitability and higher maximum losses, indicating riskier trading behavior. This suggests that traders tend to overtrade and take excessive risk during Greed periods, which reduces overall performance despite frequent winning trades.

2. **Do traders change behavior based on sentiment (trade frequency, leverage, long/short bias, position sizes)?**

Trading activity increases significantly during Fear and Extreme Fear market conditions. The number of trades is highest during Extreme Fear and lowest during Greed periods. This suggests that traders become more active during market downturns, possibly attempting to capitalize on volatility and price reversals.

Average trade size is highest during Neutral and Greed periods, while it decreases during Extreme Fear periods. This indicates that during Extreme Fear, traders execute more trades but with smaller position sizes, suggesting cautious trading behavior and risk management during highly volatile market conditions.

The long/short ratio shows that traders tend to take more long positions during Fear and Extreme Fear periods, possibly expecting market rebounds. In contrast, during Extreme Greed periods, the long/short ratio decreases, indicating more short positions, possibly due to profit-taking or expectations of market corrections.

Thus, we can conclude that trader behavior changes significantly based on market sentiment. During Fear and Extreme Fear periods, traders increase their trading activity but reduce their average position size, indicating more cautious and short-term trading strategies. In contrast, during Greed periods, traders execute fewer trades but with larger position sizes, suggesting higher risk-taking per trade. Additionally, the long/short ratio indicates that traders tend to take more long positions during Fear periods, possibly anticipating price rebounds, while short positions increase during Extreme Greed periods, possibly due to profit-taking or expectations of market corrections.

3. **Identify 2–3 segments.**

   a. **High vs low frequency traders:**

      High-frequency traders generate higher average daily PnL compared to low-frequency traders. This suggests that more active traders are better able to take advantage of market opportunities and short-term price movements. However, higher trading frequency may also expose traders to higher transaction costs and risk, which should be managed carefully.

   b. **High vs Low Position Size Traders:**

      High-frequency traders generate higher average daily PnL compared to low-frequency traders. This suggests that more active traders are better able to take advantage of market opportunities and short-term price movements. However, higher trading frequency may also expose traders to higher transaction costs and risk, which should be managed carefully.

   c. **Consistent vs Inconsistent Traders:**

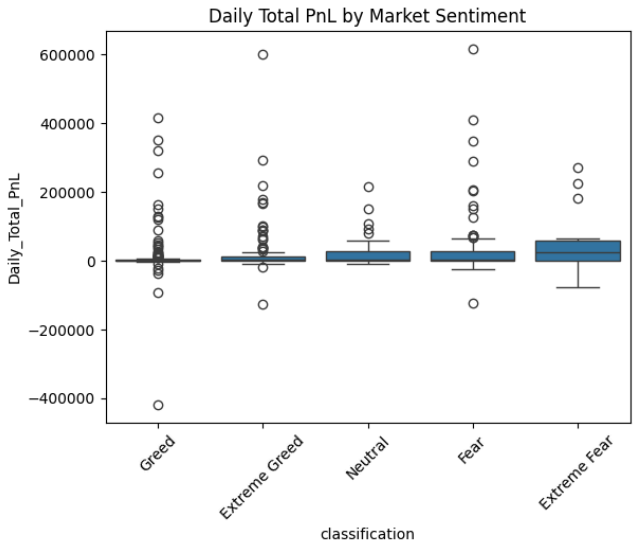
      Traders with larger average position sizes generate significantly higher profits compared to traders with smaller position sizes. This indicates that position sizing is a key driver of profitability. However, larger position sizes also increase risk exposure, meaning that risk management becomes more important for high-size traders.

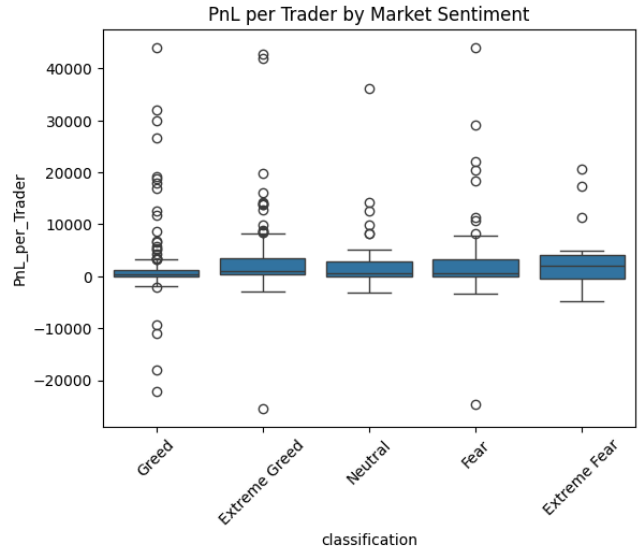
Traders can be segmented into different behavioral groups based on trading frequency, position size, and return consistency. High-frequency traders tend to generate higher daily profits compared to low-frequency traders. Traders with larger position sizes generate significantly higher profits, indicating that position sizing is a major driver of performance. Additionally, inconsistent traders, who exhibit higher PnL volatility, achieve higher average profits compared to consistent traders, suggesting a clear risk-return tradeoff in trading behavior.

Key Insights:

Insight 1: Traders Perform Better During Fear Than Greed

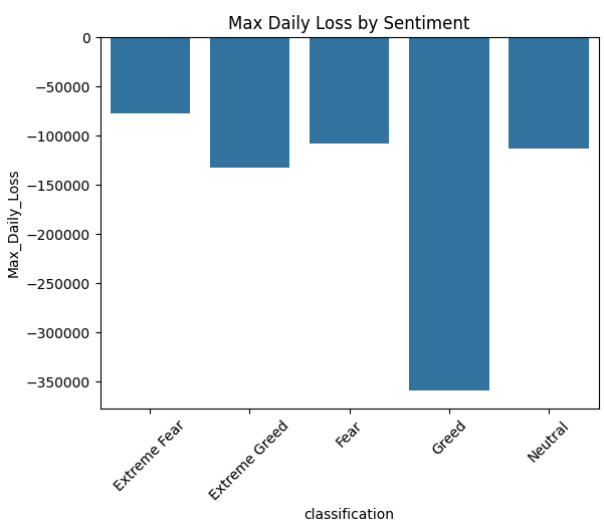
Analysis of daily PnL and PnL per trader across different market sentiment regimes shows that median profitability is higher during Fear and Extreme Fear periods compared to Greed periods. While Extreme Greed periods show occasional large profits, the overall profitability is more volatile and inconsistent. This suggests that traders perform better during fearful market conditions, possibly due to more cautious trading, better entry points, and mean-reversion opportunities.

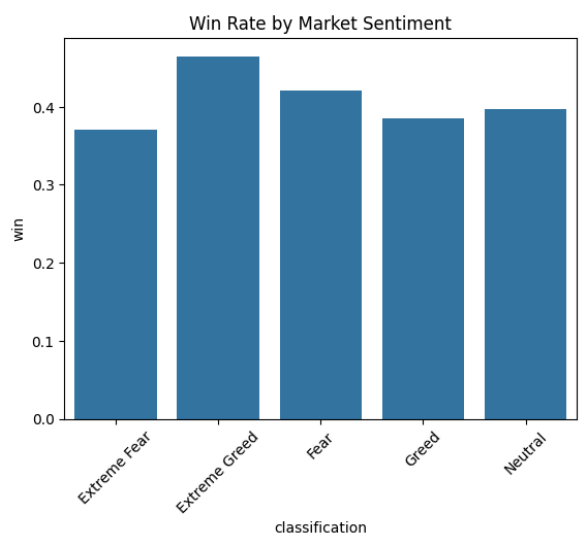


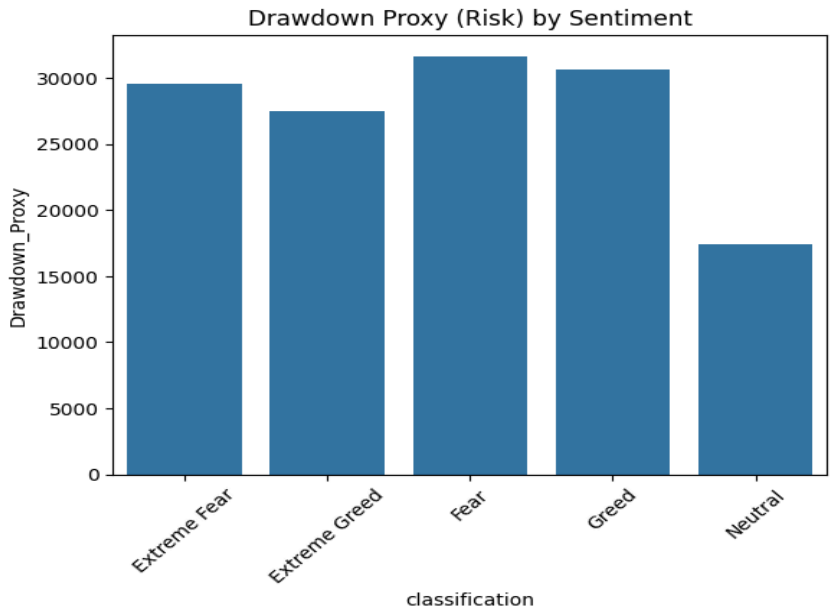


Insight 2: Higher Win Rate Does Not Mean Higher Profit

The win rate is highest during Extreme Greed periods; however, these periods do not correspond to the highest profitability. At the same time, Greed periods show the largest maximum daily losses. This indicates that traders may be making many small profitable trades during Greed markets but occasionally incur large losses that reduce overall profitability. This suggests poor risk management and overconfidence during Greed periods.

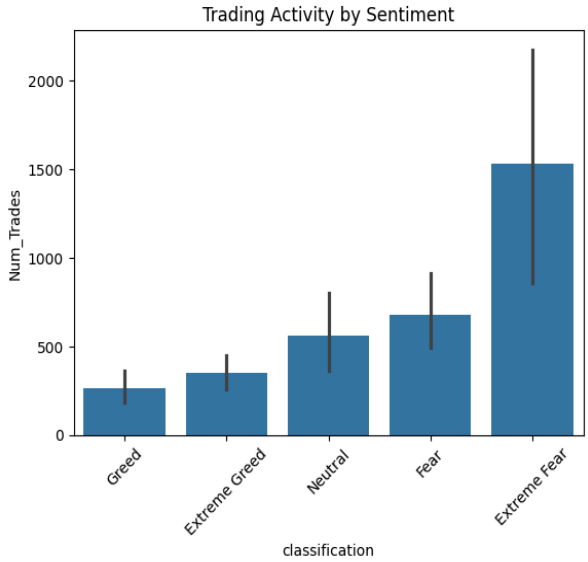


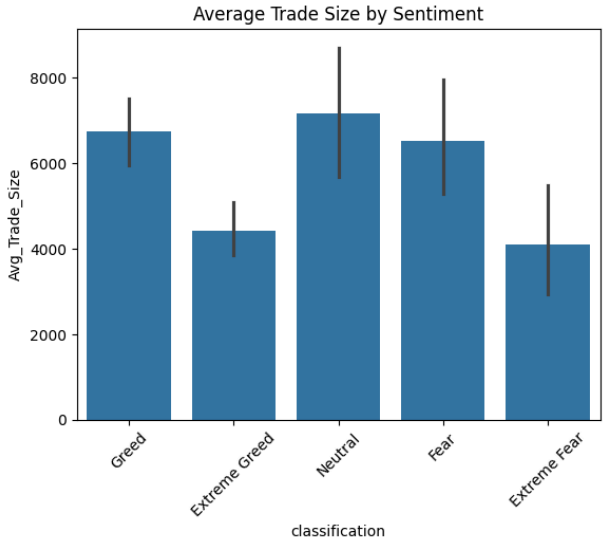




Insight 3: Traders Trade More During Fear but With Smaller Positions

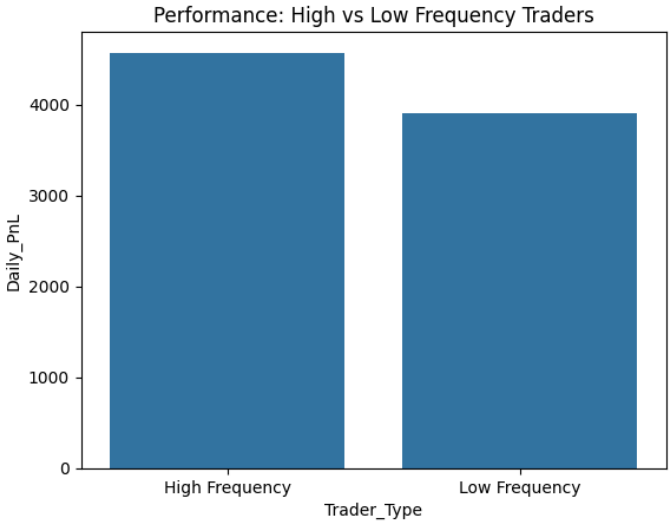
Trading activity is significantly higher during Fear and Extreme Fear periods, while average trade size is smaller during Extreme Fear periods. This indicates that traders become more active during volatile markets but reduce their position size to manage risk. In contrast, during Greed periods, traders execute fewer trades but with larger position sizes, indicating higher risk per trade.

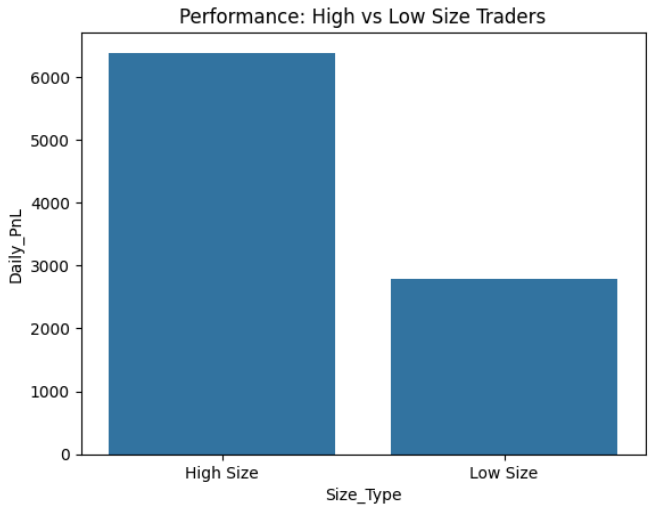


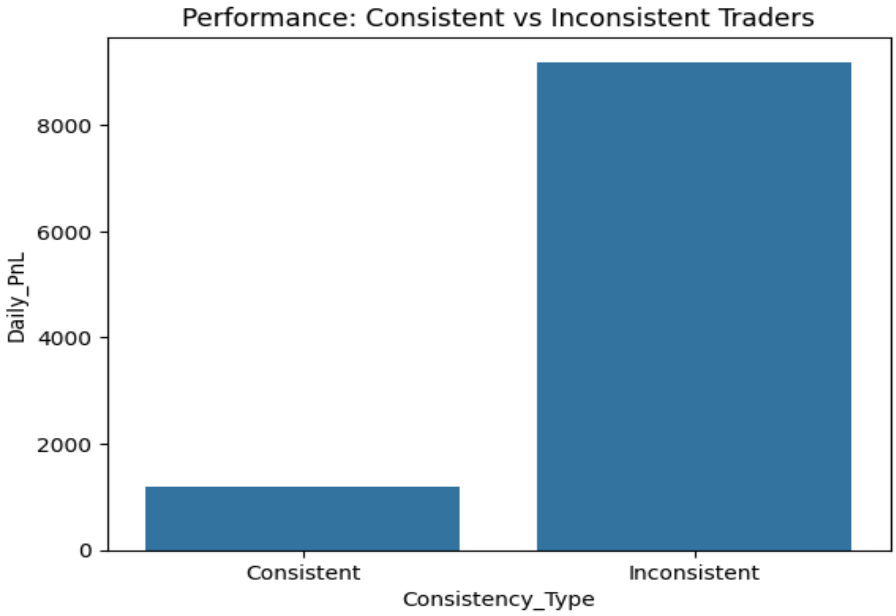


Insight 4: Profitability Is Driven by Position Size and Risk-Taking

Trader segmentation shows that high-frequency traders and high position-size traders generate higher profits. Additionally, inconsistent traders (higher PnL volatility) achieve significantly higher returns than consistent traders, indicating a strong risk-return tradeoff. Traders who take larger positions and higher risks tend to generate higher profits but also face higher volatility and drawdowns.







Strategy Recommendations:

Strategy 1: Fear Market Strategy

During Fear and Extreme Fear periods, traders should increase trading activity but maintain smaller position sizes. These periods show higher overall profitability and better trading opportunities, likely due to higher volatility and market overreactions.

Strategy 2: Greed Market Strategy

During Greed and Extreme Greed periods, traders should reduce position sizes and avoid overtrading. These periods show higher win rates but lower overall profitability and larger maximum losses, indicating poor risk-reward conditions.

Strategy 3: Risk Management Strategy

Position sizing and risk management are the most important drivers of profitability. Traders using larger position sizes generate higher profits but also face higher volatility. Therefore, traders should adjust position sizes based on market sentiment — smaller positions during Greed and larger but controlled positions during Fear.

Conclusion

This analysis demonstrates that market sentiment significantly influences trader performance and behavior. Traders tend to trade more frequently during Fear periods and take larger position sizes during Greed periods. However, profitability is higher during Fear periods, while Greed periods are associated with higher risk and larger losses. Trader segmentation further shows that profitability is strongly linked to position sizing, trading frequency, and risk-taking behavior. These findings suggest that sentiment-aware trading strategies and proper risk management can significantly improve trading performance.